

Title : **Girls, Mathematics and Science: Preparing the Next Generation to Persist, Innovate and Lead.**

Abstract: Globally, we are witnessing a profound technological transition in which Artificial Intelligence (AI), data science, automation and digital technologies are transforming education, work, research and everyday life. These technologies are characterised by sweeping newness, rapid growth, and are increasingly central to social and economic development. Yet girls and women remain underrepresented in many STEM pathways. The question for educators, researchers, policymakers, and practitioners is therefore not just how to encourage more girls in STEM, but how to create the conditions for them to persist, succeed and assume leadership within it. In this discourse, we will examine the participation and persistence of girls in Science, Technology, Engineering and Mathematics (STEM), with particular attention to Mathematics education. Mathematics in this context is viewed as a foundational language for scientific reasoning, computational thinking, data science, artificial intelligence and technological innovation. Consequently, girls' engagement with Mathematics can influence their subsequent access to scientific, technological and professional opportunities.

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